XINRU QIU, PhD

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AI-enabled Target Discovery Scientist — foundation-model interpretation, multi-omic and spatial single-cell analysis, and causal target prioritization for therapeutic discovery. *This is a one-page summary; the full CV and complete publication list are available on request — please reach out and I will send the complete version.*

Technical Skills

- **ML / Deep Learning & Foundation Models:** PyTorch, TensorFlow; single-cell foundation-model benchmarking (STATE, UCE, Geneformer, scPRINT, scGPT); DNN, XGBoost, contrastive/self-supervised learning; interpretability (SHAP)
- **Single-Cell, Spatial & Multi-Omics:** Scanpy, Seurat; spatial transcriptomics (10x Visium, Xenium), image/histology-anchored analysis; transcriptome–proteome integration; regulatory-network inference; causal inference
- **Programming & Compute:** Python, R, Bash, SQL, Git; Docker/Singularity, CI, reproducible pipelines; HPC/SLURM, GPU/CUDA, Google Cloud Platform

Selected Experience

Postdoctoral Researcher, UC Riverside (Godzik Lab) July 2023 – Present

- Designed and led an out-of-distribution robustness benchmark of five single-cell foundation models with patient-level anti-leakage cross-validation; released the open-source `scfm-benchmark` package. Sole first author.
- Built a causal-inference diagnostic (the "oracle gap"), the CausalTrapBench benchmark, and a contrastive reference ranker (CCTL) on genome-wide Perturb-seq ($\sim$2M cells), narrowing the true-gene median rank from 72 to 8. Sole first author (targeting AAAI 2027).
- Designed a verification-aware AI agent for therapeutic target validation (TPR = TNR = 1.00 on a prospective FDA-target vs. housekeeping panel).
- Multi-omics and imaging-based spatial analysis: kidney-allograft-rejection Visium study (*Frontiers in Immunology* 2025) and a carotid-atherosclerosis Xenium + mass-spectrometry-proteome integration case study (bioimage / spatial).

PhD Research Assistant, UC Riverside Jan 2019 – June 2023

- Built DNN and XGBoost models on 47,977 single-cell platelet transcriptomes to identify prognostic biomarkers (first author, *Int. J. Mol. Sci.* 2024); characterized single-cell dynamics of fatal sepsis (co-first author, *J. Leukoc. Biol.* 2021, 115+ citations); first-author, award-winning AI-for-drug-discovery review (*Biomolecules* 2024).

Earlier: Bioinformatics Intern, Illumina (DRAGEN NGS/pharmacogenomics, 2022) · Clinical Genomics Scientist, Vantari Genetics (NGS, ACMG variant classification, CAP/CLIA, 2015–2018).

Education

PhD, Genetics, Genomics & Bioinformatics, UC Riverside (2023) · MS, Pharmaceutical Sciences, USC (2015) · BS, Pharmacy, Jilin University (2013)

Highlights

20+ peer-reviewed publications · 400+ citations · h-index 10 · 4 first-author papers · 200+ peer reviews across 130+ manuscripts · reviewer for ICLR, ICML, NeurIPS · *Biomolecules* Best Paper Award · Topic Editor, *Frontiers in Bioinformatics*.

Full CV available on request — contact me and I will send the complete version.